

Field Theory

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1. FIELD EXTENSIONS

1.1. INTRODUCTION

We begin by studying a fundamental invariant of fields. Recall that given any field F there is a natural ring morphism¹ $\varphi : \mathbb{Z} \rightarrow F$. Since the image of φ is an integral domain, we get that $\ker \varphi$ must be a prime ideal of $\mathbb{Z}$, that is $\ker \varphi$ is either (0) or (p) for some prime p . This leads us to our first definition;

Definition 1.1. The **characteristic** of a field F , denoted by $\text{char } F$, is defined to be the nonnegative principle generator of $\ker \varphi$ where $\varphi : \mathbb{Z} \rightarrow F$ is the natural ring morphism.

In the case when $\text{char } F$ is 0, notice that φ is an embedding of $\mathbb{Z}$ in the field F . Indeed, we may extend this to an embedding of its *quotient field* $\mathbb{Q}$ in F . Further, when $\text{char } F$ is a prime p , then by the first isomorphism theorem, we get an embedding of $\mathbb{F}_p$ in F . It is easily checked that this is the smallest field contained² in F .

Infact we define the smallest field contained in F to be the *prime subfield* of F . From our discussion it is clear that the prime subfield of F must be isomorphic to either $\mathbb{Q}$ or $\mathbb{F}_p$.

Definition 1.2. If F is a subfield of a field E , then E is called an **extension** of F . We denote this by E/F .

Recall that the only ideals of a field F are the trivial ideal and itself. It follows that every ring morphism of fields is injective. Given a ring homomorphism of fields $\varphi : F \rightarrow E$, we may view E as an extension of F by identifying F with its isomorphic copy in E .

Moreover, the characteristic of a field is invariant under homomorphisms, hence every field is the extension of its prime subfield.

Given an extension E/F , we may view E as a F -vector space.

Definition 1.3. Let E/F be an extension. We define the **degree** of the extension as $[E : F] = \dim_F E$.

Theorem 1.4. Given a tower of fields $E/F/K$, we have the identity

$$[E : K] = [E : F][F : K].$$

Proof. Let $[E : F] = m$ and $[F : K] = n$. Let $\{\alpha_i\}_{i=1}^m$ be a basis of E/F and $\{\beta_j\}_{j=1}^n$ be a basis of F/K . Our claim follows by showing that $\{\alpha_i \beta_j\}_{ij}$ is a basis of E/K . $\square$

We give a few more definitions. Let E/F be an extension.

¹We require the identity to map to the identity

²“contained” here actually means F contains an isomorphic subfield

Definition 1.5. Let $S \subseteq E$. We denote the smallest subfield of E that contains both F and S by $F(S)$. If S is finite, then $F(S)$ is said to be finitely generated.

Definition 1.6. An extension E/F is finite if $[E : F] < \infty$.

Definition 1.7. An element $\alpha \in E$ is **algebraic** over F if it is the root of some nonzero polynomial in $F[x]$. If α does not satisfy any such polynomial then it is called **transcendental**. The extension E/F is algebraic if all elements of E are algebraic over F .

If $\alpha \in E$ then we get a F -homomorphism $\varphi : F[x] \rightarrow F(\alpha)$ given by $x \mapsto \alpha$. Note that $\ker \varphi$ is the set of all polynomial in $F[x]$ which have α as a root. Since $F(\alpha)$ is an integral domain, $\ker \varphi$ must be a prime ideal. Further, since $F[x]$ is a PID, $\ker \varphi$ is either (0) or (p) for some irreducible $p \in F[x]$.

In the case when $\ker \varphi$ is trivial, we get that α is transcendental. This also gives that $F[x]$ is embedded in $F(\alpha)$. We may extend this to an embedding of the quotient field of $F[x]$ (denoted $F(x)$) in $F(\alpha)$. Since $F(\alpha)$ is the smallest field containing both α and F , we get $F(x) \cong F(\alpha)$.

In the case when $\ker \varphi = (p)$, then α is algebraic over F . Observe that $\text{Im } \varphi \cong F[x]/(p)$ is a field (since the ideal generated by an irreducible is maximal in a PID) which contains both F and α ; hence $F(\alpha) = \text{Im } \varphi$. This polynomial is indeed quite special, we give it a name.

Definition 1.8. Let E/F be an extension and let $\alpha \in E$ be algebraic over F . The minimal polynomial of α in F , denoted $\text{Irr}(F, \alpha)$, is the unique monic irreducible $p \in F[x]$ which has α as a root³.

We must check the uniqueness of $\text{Irr}(F, \alpha)$ before we proceed any further. From the above discussion, we get that $\ker \varphi = (p)$ for some irreducible $p \in F[x]$, since α is algebraic. Assume, without loss of generality, that p is monic. We show that $p = \text{Irr}(F, \alpha)$. Suppose g is another monic irreducible with α as a root. Then $g \in \ker \varphi = (p)$, so $g \mid p$ i.e. they are associates. But since g is also monic, they must be equal.

Theorem 1.9. If E/F is an extension and $\alpha \in E$ is algebraic over F then we have $F[x]/(\text{Irr}(F, \alpha)) \cong F(\alpha)$ and $[F(\alpha) : F] = \deg(\text{Irr}(F, \alpha))$.

Proof. The first part has been proved in the discussion above. Let $p = \text{Irr}(F, \alpha)$ and let $p(x) = a_0 + a_1x + \dots + a_nx^n$ for some $a_i \in F$ with $a_n \neq 0$. We claim that $\mathcal{B} = \{1, \alpha, \dots, \alpha^{n-1}\}$ forms a basis of $F(\alpha)/F$. Indeed they are linearly independent since any nontrivial combination would give a polynomial of smaller degree than p which has α as a root, contradicting the irreducibility of p . Now, to show $\mathcal{B}$ spans $F(\alpha)$, observe that $\alpha^n = -a_{n-1}\alpha^{n-1} - \dots - a_0$ is in $\text{span } \mathcal{B}$. We can inductively show that $\alpha^m \in \text{span } \mathcal{B}$ for all $m \in \mathbb{N}$. Since $F[\alpha] = F(\alpha)$, our result follows. $\square$

Now we give a few theorems on finite, finitely-generated and algebraic extensions and how they relate to each other.

Theorem 1.10. A finite extension is finitely-generated. The converse is false.

Proof. Suppose E/F is finite. Let $\alpha_1, \dots, \alpha_n$ be a basis of E/F . Clearly $E = F(\alpha_1, \dots, \alpha_n)$. Conversely, consider $F(x)/F$ where x is a formal variable. The set $\{x^n\}_{n=1}^\infty$ is linearly independent over F , our result follows. $\square$

Theorem 1.11. All finite extensions are algebraic.

Proof. Let E/F be a finite extension, say of degree n . Take any $\alpha \in E$ then the set $\{1, \alpha, \dots, \alpha^n\}$ is linearly dependent over F . Hence, there is a nonzero polynomial that α satisfies showing that α is algebraic over F . $\square$

³when viewed as a polynomial in $E[x]$

Theorem 1.12. Let E/F be an extension. If $\alpha, \beta \in E$ are algebraic over F then so is $\alpha + \beta$.

Proof. Observe that $[F(\alpha, \beta) : F(\alpha)] \leq [F(\beta) : F]$ since $\text{Irr}(F, \beta)$ will still have β as a root in $F(\alpha)$, hence the degree of the former extension cannot increase. By 1.4 and 1.11, $[F(\alpha, \beta) : F] \leq [F(\beta) : F][F(\alpha) : F]$ is finite hence algebraic. Since $\alpha + \beta \in F(\alpha, \beta)$, we get that $\alpha + \beta$ is algebraic over F . $\square$

Corollary 1.13. It follows that if $\alpha, \beta \in E$ are algebraic over F then $\alpha \cdot \beta$ is also algebraic. Furthermore, the algebraic elements of E hence form a field (infact, a subextension of E/F).

Theorem 1.14. Let $E/F/K$ be a tower of fields where E/F and F/K are algebraic extensions. Then E/K is algebraic.

Proof. Pick any $\alpha \in E$. Then α satisfies some polynomial $p(x) = a_0 + \cdots + a_n x^n \in F[x]$. Let $S = \{a_0, \dots, a_n\} \subset F$ be the set of coefficients of p . Observe that $[K(S) : K]$ is finite since

$$[K(S) : K] \leq \prod_{i=0}^n [K(a_i) : K].$$

Moreover, α is clearly algebraic over $K(S)$. In particular, by 1.4 we have

$$[K(S, \alpha) : K] = [K(S, \alpha) : K(S)][K(S) : K] < \infty.$$

By 1.11, $K(S, \alpha)/K$ is algebraic, our claim follows. $\square$

2. UNSORTED

2.1. ALGEBRAIC CLOSURE

Remark 2.1. Let F/K be an extension. Given elements $\alpha_1, \dots, \alpha_n \in F$, we get that

$$K[\alpha_1, \dots, \alpha_n] \subset K(\alpha_1, \dots, \alpha_n).$$

Infact, because $k[\alpha_1, \dots, \alpha_n]$ is a field. We prove this inductively.

Remark 2.2. A field F is algebraically closed if every nonconstant polynomial over F has a root. The algebraic closure of a field E is a field $\bar{E}$ such that $\bar{E}/E$ is algebraic and $\bar{E}$ is algebraically closed.

Remark 2.3. Given a field F , there is an algebraically closed field E containing F .

Proof. We construct a field E_1 in which all polynomial of $F[x]$ with degree ≥ 1 have a root. For each nonconstant polynomial $f \in F[x]$, associate a symbol x_f . Let S be the set of all these symbols. Let I denote the ideal generated by $f(x_f)$ for all f in $F[S]$. We claim that I is a proper ideal. Suppose not, then there are $f_1, \dots, f_n$ (resp. $x_1, \dots, x_n$) and polynomials $g_1, \dots, g_n \in K[S]$ such that

$$g_1 f_1(x_1) + \cdots + g_n f_n(x_n) = 1.$$

Let N be such that the polynomials g_i contain only the variables $x_1, \dots, x_N$. Then our relation becomes

$$\sum_{i=1}^n g_i(x_1, \dots, x_N) f_i(x_i) = 1.$$

Let F' be the finite extension in which f_i have roots, say α_i for $i = 1, \dots, n$. Putting $x_i = \alpha_i$ for $i \leq n$ and $x_i = 0$ otherwise; we get $0 = 1$, a contradiction.

Since all ideals are contained in a maximal ideal, let $\mathfrak{m} \subset F[S]$ be a maximal ideal containing I . Then $F[S]/\mathfrak{m}$ is a field. Observe that all nonconstant $f \in F[S]$ have roots in $F[S]/\mathfrak{m}$, namely the image of x_f .

TODO $\square$

2.2. SPLITTING FIELD AND NORMAL EXTENSIONS

Remark 2.4. (Splitting Field) Let F be a field and $f \in F[x]$ of degree ≥ 1 . A splitting field of f is an extension E/F such that

$$f(x) = c(x - \alpha_1) \cdots (x - \alpha_n)$$

in $E[x]$. Further, $E = F(\alpha_1, \dots, \alpha_n)$.

All splitting fields of f are F -isomorphic.

Remark 2.5. (Normal Extension) Let E/F be an algebraic extension. The following are equivalent and define a normal extension

1. Every embedding $\sigma : E \rightarrow \Omega$, arbitrary Ω , over F is an automorphism of E .
2. E is the splitting field of a family of polynomials in $F[x]$.
3. The minimal polynomials of each $\alpha \in E$ splits into linear factors in $E[x]$.
4. Every irreducible polynomial of $F[x]$ which has a root in E splits into linear factors in $E[x]$.

Let $E/F/K$ be a tower. If E/F and F/K are normal, that does not mean that E/K is normal. Consider $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$, this is not normal since $x^4 - 2$ does not have all its roots in it. We can view it as $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ and degree two extensions are normal.

Remark 2.6. Let E/F be algebraic and $\sigma : F \rightarrow L$ be an embedding of F into an algebraically closed field L . Then there is an extension of σ to an embedding of E into L .

2.3. SEPARABLE EXTENSIONS

Remark 2.7. (Separable degree) Let E/F be an algebraic extension. Let σ and τ be embeddings of F into algebraically closed field L and L' respectively. We assume that $L/\sigma F$ and $L'/\tau F$ are algebraic. Consider the sets S_σ of embeddings of E into L over σ , define S_τ analogously. Then there is an isomorphism $\lambda : L \rightarrow L'$ extending $\tau \circ \sigma^{-1}$ applied to σF .

$$\begin{array}{ccccc}
 & & L' & \xleftarrow{\lambda} & L \\
 & \swarrow \tau^* & & & \searrow \sigma^* \\
 & & E & & \\
 \downarrow & & \downarrow & & \downarrow \\
 \tau F & \xleftarrow{\tau} & F & \xrightarrow{\sigma} & \sigma F
 \end{array}$$

Observe that $\lambda_* : S_\sigma \rightarrow S_\tau$ is a bijection. Hence, the cardinality of S_σ (or S_τ) depends only on the extension E/F . Define the separable degree $[E : F]_s$ as the to be S_σ for some $\sigma : F \rightarrow L$.

(Multiplicativity of separable degree) Let $E/F/K$ be a tower. Then

$$[E : K]_s = [E : F]_s [F : K]_s.$$

If E/K is finite, then $[E : K]_s$ is finite and

$$[E : K]_s \leq [E : K].$$

Remark 2.8. Let E/F be an extension and $\alpha \in E$ be algebraic with $p(x) = \text{Irr}(F, \alpha)$. Then α is a simple root of p if $p'(\alpha) \neq 0$.

Let E/F be an extension and let $\alpha \in E$ be algebraic over F . Then α is **separable** over F if α is a simple root of $\text{Irr}(F, \alpha)$.

Let F be a field. An irreducible polynomial P over F is separable if and only if P has pairwise distinct roots in an algebraic closure of F . Observe that a polynomial $P \in F[x]$ is separable if $(P, P') = (1)$.

Remark 2.9. Let F be a field. Let $P \in F[x]$ be irreducible. Then there is the dichotomy

1. P and P' are relatively prime, or
2. P' is the zero polynomial.

Case (1) occurs when $\text{char } F = 0$. On the other hand, if $\text{char } F = p$ then both case (1) and (2) can occur. Moreover, if case (2) occurs then $P(x) = Q(x^p)$ for some irreducible $Q \in F[x]$.

Proof. Observe that $\deg P' < \deg P$. Hence, if $P' \neq 0$, then $(P, P') = (1)$ since P is irreducible.

Assume we are in case (2) and let $P = a_0 + \cdots + a_n x^n$, then $P' = a_1 + \cdots + n a_n x^{n-1}$. If $\text{char } F = 0$, then $P' = 0$ forces $P = a_1$, contradicting the irreducibility of P . If $\text{char } F = p$, then $P' = 0$ forces $p \mid a_i$ whenever $p \nmid i$, i.e. $a_i = 0$. Hence $P(x) = Q(x^p)$ for some $Q \in F[x]$. Observe that Q is irreducible since a factorisation of Q would give a factorisation for P . $\square$

Remark 2.10. (Separability) Let E/F be a finite extension. The following are equivalent and define a separable extension

1. $[E : F]_s = [E : F]$.
2. Every element of E is separable over F .

Remark 2.11. Let α be algebraic over F . Then there is $\nu \geq 0$ such that every root of $\text{Irr}(F, \alpha)$ has multiplicity p^ν and

$$[F(\alpha) : F] = p^\nu [F(\alpha) : F]_s.$$

Remark 2.12. (Primitive Element Theorem) Every finite separable extension E/F is simple, i.e. there is $\alpha \in E$ such that $E = F(\alpha)$.

2.4. THE FROBENIUS ENDOMORPHISM

A field F is **perfect** if every algebraic extension is separable.

Remark 2.13. (Frobenius endomorphism) Let F be a field of prime characteristic p . Define the endomorphism $\varphi : F \rightarrow F$ given by $x \mapsto x^p$. This is indeed an endomorphism since $p \mid \binom{p}{r}$ for $0 < r < p$. The image of φ is denoted by F^p .

Note that this is not necessarily surjective. Consider $\mathbb{F}_p(t)$ where t is transcendental. Then the image of φ does not contain t . If it did, then there would be $f(t), g(t) \in \mathbb{F}_p[t]$ such that $f^p(t)/g^p(t) = t$. But the degree of $f^p(t)/g^p(t)$ is $p(\deg f - \deg g)$. Since p divides the degree, it cannot be 1, which is the degree of t .

Remark 2.14. The field F is perfect if and only if either $\text{char } F = 0$ or $\text{char } F = p$ and $F^p = F$.

Proof. Let E/F be algebraic and let $\alpha \in E$ with $P(x) = \text{Irr}(F, \alpha)$. If $\text{char } F = 0$, then $(P, P') = (1)$ showing that P has no repeated roots, so α is separable.

If $\text{char } F = p$ and $F^p = F$. If $(P, P') = (1)$, then our result follows by repeating the previous argument. If $P' = 0$, then $P(x) = Q(x^p)$ for some irreducible $Q(x) = a_0 + \cdots + a_n x^n \in F[x]$. Since the Frobenius map is an isomorphism, we get $b_i \in F$ such that $b_i^p = a_i$. Let $\tilde{Q}(x) = b_0 + \cdots + b_n x^n$. It follows that $\tilde{Q}^p(x) = P(x)$, contradicting the irreducibility of P .

Conversely, suppose $\text{char } F = p$ and F is perfect. For any $\alpha \in F$, consider $x^p - \alpha \in F[x]$. If this polynomial does not have a root in F ,

TODO

$\square$

It follows that all finite fields are perfect.

2.5. GALOIS STUFF

Remark 2.15. Let $G \leq \text{Aut}(F)$. Then the fixed field of G is

$$F^G = \{x \in F \mid x^\sigma = x \forall \sigma \in G\}.$$

This is a field. Observe that the automorphisms of prime subfields are trivial, hence F^G must also contain the same prime subfield as F .

Remark 2.16. An extension E/F is called **Galois** if it is normal and separable.

The group of automorphisms of an extension E/F which fixes F is denoted by $\text{Aut}_F(E)$, in the case when E/F is Galois, it is denoted by $\text{Gal}(E/F)$.

Remark 2.17. Let E/F be Galois and let $G = \text{Gal}(E/F)$. Then $F = E^G$.

Proof. Clearly $F \subseteq E^G$, we need only prove the reverse containment. Let $\alpha \in E^G$. Since $F(\alpha)$ is algebraic over F and $F \hookrightarrow \bar{E}$, by 2.6 we get an embedding $\sigma : F(\alpha) \rightarrow \bar{E}$. Again, since E is algebraic over $F(\alpha)$, using 2.6 we can extend σ to an embedding $\sigma : E \rightarrow \bar{E}$. Since E/F is normal, σ is an automorphism of E over F , so $\sigma \in G$. Hence σ fixes α , and since σ is arbitrary we get

$$[F(\alpha) : F]_s = 1.$$

As E/F is separable, α is separable and hence $[F(\alpha) : F] = [F(\alpha) : F]_s = 1$. Therefore, $F = F(\alpha)$ and $\alpha \in F$. $\square$

Remark 2.18. Let $E/F/K$ be a tower with E/K Galois then E/F is also Galois. Further, the map

$$F \mapsto \text{Gal}(E/F)$$

from the set of intermediate fields to the subgroups of $\text{Gal}(E/K)$ is injective.

Proof. It follows that E/F is separable and normal. [Lan02] p.192 [TODO]. $\square$

Remark 2.19. Let E/F be algebraic. Let $n \geq 1$ be such that every element of E has degree $\leq n$ over F . Then $[E : F] \leq n$.

Proof. Let $\alpha \in E$ be such that $[F(\alpha) : F] = m$ is maximal. We contend that $F(\alpha) = E$. Suppose not, then there is $\beta \notin F(\alpha)$. By the Primitive Element Theorem, there is $\gamma \in F(\alpha, \beta)$ such that $F(\alpha, \beta) = F(\gamma)$. But from the tower

$$F \subset F(\alpha) \subset F(\alpha, \beta),$$

we infer that $[F(\alpha, \beta) : F] > m$, hence γ has degree $> m$ over F , a contradiction. $\square$

Remark 2.20. Let F be a field and let $G \leq \text{Aut}(F)$ be finite. Then F/F^G is Galois and

$$[F : F^G] = |G|.$$

Proof. Let $\alpha \in F$ and let $\{\alpha_i\}_{i=1}^r$ be the orbit of α under action of G on F . Then for any $\sigma \in G$, we have $\{\sigma\alpha_i\}_{i=1}^r$ is a permutation of $\{\alpha_i\}_{i=1}^r$. Consider the polynomial

$$f(x) = \prod_{i=1}^r (x - \alpha_i),$$

observe that $f(\alpha) = 0$ and $f^\sigma = f$ for all $\sigma \in G$. This shows that all the coefficients of f are fixed by G , hence $f \in K^G[x]$. Moreover, $\text{Irr}(F^G, \alpha)$ divides f , hence α is separable since the factors

are distinct. Furthermore, the extension is normal since the minimal polynomial of every element splits in $F[x]$. Hence the extension F/F^G is Galois.

We now prove our second assertion. Notice that every element $\alpha \in E$ has degree $\leq |G|$ in F , hence by 2.19, we have $[K : K^G] \leq |G|$. Now, $G \leq \text{Gal}(K/K^G)$ and $\text{Gal}(K/K^G) \leq [K : K^G]_s = [K : K^G]$ since the extension is separable. Hence $[K : K^G] = |G|$ and $G = \text{Gal}(K/K^G)$. $\square$

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